CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

May – 2012

EC 201 Digital Electronics & Logic Design

Date: 01.05.2012, Tuesday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Find the 10's complement of $(935)_{11}$. [01]

(b) Convert Decimal 225.225 to binary, octal and hexadecimal. [03]

(c) Why NAND and NOR gates are known as universal gates? [03]

Q - 2 (a) Express the Boolean function F=XY+X'Z in a product of max term form. [03]

(b) Write down the design procedure steps for designing any combinational circuit. [05]

(c) Design a combinational circuit that converts a decimal digit from 8, 4, -2, -1 code to BCD. [06]

OR

Q - 2 (a) Simplify the following boolean functions to minimum number of literals. [04]

(i) X+X'Y

(ii) XY+X'Z+YZ

(b) Show how a Full Adder can be converted to a full subtractor with the addition of one inverter circuit. [04]

(c) Design a combinational circuit that converts a decimal digit from BCD to excess-3 [06]

Q - 3 (a) Simplify the function $F(A,B,C,D,E,F,G)=\sum(20,28,38,39,52,60,102,103,127)$ with either K-map or Tabulation method. [08]

(b) Construct a 6×64 Decoder with 3×8 Decoders. [06]

OR

Q - 3 (a) Give three possible ways to express the function: [07]

$$F=A'B'D'+AB'CD'+A'BD+ABC'D$$

(b) Implement function [07]

$$F(A,B,C,D) = \sum(1,5,8,10,12,13,15)$$

with 8×1 Multiplexer.

SECTION – II

Q - 4 (a) What is the difference between Combinational circuit and Sequential circuit? [02]

(b) Explain edge triggered D-Flip flop. [05]

Q - 5 (a) Design a counter with the following binary sequence 0,1,2,4,5,6 and repeat. Use J-K flip flop. [07]

(b) Design a serial adder using a sequential-logic procedure. [07]

OR

Q - 5 (a) Design 4 bit Binary Ripple Counter. [07]

(b) Implement functions [07]

$$F1(A,B,C,D) = \sum(1,2,4,6,8,9,12,15)$$

and

$$F2(A,B,C,D) = \sum(3,6,7,12,13,15)$$

with ROM.

Q - 6 (a) Design 4 bit binary Synchronous Counter. [07]

(b) Explain Master slave flip flop. [07]

OR

Q - 6 (a) Explain Johnson Counter. [08]

(b) Explain [06]

(1) Bidirectional Shift register

(2) Memory unit

****